Grace — BLIND pre-registration: alpha-conjugate binding ladder (Frontier C)

2026-08-03 ~14:00. Committed BEFORE Elie's binding-energy toy lands, so his energies are a TEST not a fit.

What I commit blind (target-innocent — pure geometry)

Bond-count ladder from tetrahedral close-packing of n alpha-tetrahedra (alpha = 4-nucleon tetrahedron, $C(4,2)=C_2$):

n	nucleus	P(n) bonds	geometry
1	^{4}He	0	single tetra
2	^{8}Be	1	2 in contact
3	^{12}C	3	triangle, C(3,2)=3
4	^{16}O	6	tetrahedron of alphas, $C(4,2)=C_2=6$
5	^{20}Ne	9	+face sphere, +3
6	^{24}Mg	12	+face sphere, $+3 = 2C_2$

P(n) = {0,1,3,6,9,12}: first four = triangular numbers T_{n-1}; then +3/sphere (3D close-packing saturates at 3 new contacts per added tetra).

Committed: (1) the ladder P(n); (2) the form $B = n \cdot B\_\alpha + P \cdot \varepsilon$ (6 points / 2 params → overdetermined, 4 dof); (3) the self-similar C_2 — ^{16}O's 6 alpha-bonds = the alpha's own 6 nucleon-bonds, one shape at two scales.

Honesty

This linear-in-bonds law IS the known Hafstad–Teller (1938) alpha-particle model — I am NOT claiming to invent it. The BST contribution: the bond counts fall out of the SAME tetrahedron/C_2 I nailed for the alpha block (native, not imported), and I'm pre-registering it under K1133 discipline.

The discipline (K1133 rides along — the κ lesson)

- COUNT P(n): forced + target-innocent. Good.
- STRENGTHS B_α, ε: the load-bearing question. $B\_\alpha=13 \cdot B\_d$ already a BST ratio (28.9 vs 28.3, ~2%). ε forced→Derived; ε fitted→PD (count forced/strength fit), same shape as κ.
- KEY difference from κ: 6 data points, 2 params. Even if ε ends up fitted, the STRUCTURE (linear-in-bonds) is overdetermined — real evidence, not a one-number coincidence. This frontier can reach Derived where κ couldn't, IF ε is geometry-forced.

Sanity check only (Elie owns the real fit)

Least-squares $B = n \cdot B\_\alpha + P \cdot \varepsilon$ over the 6 nuclei: $B\_\alpha \approx 27.75$ MeV, $\varepsilon \approx 2.60$ MeV, max resid 1.6 MeV (~1.5%). Linearity holds. Elie's job: derive ε from geometry (blind) → the make-or-break, exactly as κ was.